

Figure 12.7: **(a)** Transition mutations **(b)** transversion mutations

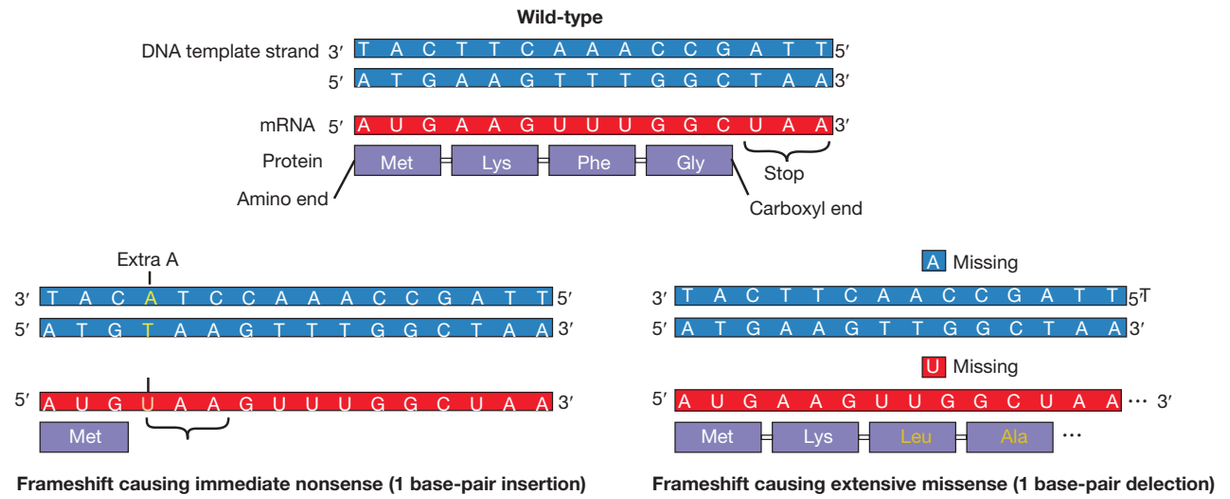


Figure 12.8: Frameshift mutations

Point mutations are single base changes, that do not affect the reading frame, that is, the mutation only makes a single change in a single codon, and everything else is undisturbed.

Mutations can also be defined according to their effects on amino acid sequences in proteins. They are

1. A missense mutation (Figure 12.9a) is a gene mutation in which a base – pair change in the DNA changes a codon in an mRNA so that a different amino acid is inserted into the polypeptide.
2. A neutral mutation (Figure 12.9b) is a subset of missense mutations in which the new codon codes for a different amino acid that is chemically equivalent to the original and therefore does not affect the proteins function. Consequently, the phenotype does not change.
3. A silent mutation (Figure 12.9c) is also a subset of missense mutations that occurs when a base – pair change in a gene alters a codon in the mRNA such that the same amino acid is inserted in the protein. In this case, the protein obviously has a wild type function.
4. A nonsense mutation (Figure 12.9d) is a gene mutation in which a base – pair change in the DNA, changes a codon in an mRNA to a stop (nonsense) codon (UAG, UAA or UGA). Nonsense mutation cause premature chain termination so instead of complete polypeptides, shorter than

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